25/8/26

COAL LAB

Lab Tasks.

1. Describe the function of each

- Accumulator Register (A, AX, EAX, RAX)
The Accumulator register is mainly used to store operations and results of arithmetic and logical operations. It is frequently used by the CPU for calculations.
AX - 16 bit accumulator
EAX - 32 bit accumulator
RAX - 64 bit accumulator

- Base Register (B, BX, EBX, RBX)
The Base register is commonly used to store a base address for accessing data in memory. It can also be used for general-purpose calculations
BX - 16 bit
EBX - 32 bit
RBX - 64 bit

3. Count Register (C, CX, ECX, RCX)
The count register is mainly used as a counter for loops and repeated operations. It also used by shift and rotate instructions.

4. Data Register (D, DX, EDX, RDX)
The Data register is used to store data during arithmetic operations and

Scanned with CamScanner

is especially important in multiplication, division, and I/O operations.

Q2. What are Index Registers? briefly explain with examples.

Index registers are CPU registers used to hold an offset or index for accessing elements in memory, especially arrays and strings.

The main index registers in x86 are:

SI (Source Index): used mainly for source data.

DI (Destination Index): used mainly for destination data.

Example:

```
MOV SI, 1000h
MOV AX, [SI]
```

Here, SI contains the offset/address used to access data from memory.

Q3: Briefly describe the functions of these bits.

1. Overflow flag (OF):
The overflow flag is set when a signed arithmetic operation produces a result that is too large or too small to fit in the destination register.

2. Zero Flag (ZF):
The zero flag is set when the result of an operation is zero.

Scanned with CamScanner

3. Carry Flag (CF):
The carry flag is set when an arithmetic operation produces a carry out of the most significant bit. It is mainly used for unsigned arithmetic.

4. Parity flag (PF):
The parity flag indicates whether the result contains an even number of 1 bits in its least significant byte

5. Direction Flag (DF).
The Direction Flag controls the direction of string operations.
DF = 0 → string operation move forward
DF = 1 → string operation move backward.

Scanned with CamScanner